

RESEARCH

Open Access



Pregnancy and survival-related outcomes of uveal melanoma treated with brachytherapy in women of reproductive age

Hao-Tian Wu¹, Li Dong¹, Rui-Heng Zhang¹, Wen-Da Zhou¹, He-Yan Li¹, Wen-Bin Wei^{1*} and Yue-Ming Liu^{1*}

Abstract

Background To examine if pregnancy affects the prognosis of uveal melanoma (UM) patients undergoing plaque brachytherapy (PBT) and to assess if PBT has any subsequent impact on pregnancy outcomes.

Methods A retrospective, single-center study was carried out at Beijing Tongren Hospital, focusing on women of childbearing age diagnosed with UM and treated with iodine-125 plaque brachytherapy. Both the outcomes of pregnancies and the health status of the fetuses were monitored. Survival analyses were conducted using the Kaplan-Meier method, with endpoints being metastasis and death.

Results A total of 13 patients who had full-term pregnancies and 96 non-pregnant women matched by age and tumor size were included. The mean follow-up time was 67.0 ± 27.7 months (median: 66.0 months, range: 21.0 to 116.0 months). In the pregnant group, two patients developed metastases, one of whom died shortly after delivery; local recurrence of UM occurred in 2 patients after or during delivery, and 2 other patients developed secondary glaucoma due to radiation retinopathy. None of the other pregnant patients reported any signs of disease progression. In the control group, 18 metastasis cases including 12 deaths were documented. Pregnant patients and matched control subjects showed no statistical difference in both Metastasis-free survival (hazard ratio (HR): 0.66, 95% confidence interval (CI): 0.15–2.86; $P=0.576$) and overall survival (HR: 0.48, 95% CI: 0.06–3.66; $P=0.464$). All pregnant patients carried the pregnancy to term and delivered healthy children with no report of placental or infant metastases to date.

Conclusion Pregnancy does not appear to negatively impact the prognosis of UM patients undergoing PBT. PBT showed no observable detriment to maternal fertility and exhibited no teratogenic effects on the fetus. However, the long-term implications of PBT on pregnancy remain uncertain, necessitating additional, prolonged follow-up studies.

Keywords Pregnancy, Uveal melanoma, Plaque brachytherapy

Wen-Bin Wei and Yue-Ming Liu shared the co-correspondence authors.

*Correspondence:

Wen-Bin Wei
weiwenbintr@163.com
Yue-Ming Liu
liuyueming2005@163.com

¹Beijing Key Laboratory of Intraocular Tumor Diagnosis and Treatment, Beijing Ophthalmology & Visual Sciences Key Lab, Medical Artificial Intelligence Research and Verification Key Laboratory of the Ministry of Industry and Information Technology, Beijing Tongren Hospital, Beijing Tongren Eye Center, Capital Medical University, 1 Dong Jiao Min Lane., Beijing 100730, China



© The Author(s) 2024. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Introduction

Uveal melanoma (UM) is the most common primary intraocular malignant tumor in adults [1], with a 5-year survival rate ranging from 25–66% [2]. A significant proportion of UM cases may culminate in compromised vision or even necessitate enucleation [3]. Plaque radiotherapy is a treatment in which a curvilinear radioactive plaque is sutured to the sclera, precisely over the tumor, to deliver trans-scleral radiation to the UM. When administered by experienced clinicians and with judicious patient selection, this therapy can achieve survival outcomes mirroring those from alternative treatments such as enucleation [4, 5]. Furthermore, it results in globe salvage for approximately 95% of patients, often preserving functional vision [3]. Consequently, this treatment has emerged as the primary therapeutic choice for most UM patients at our institution.

While melanoma is predominantly diagnosed in older male individuals [6, 7], a notable cohort of women within the childbearing age (15–45 years) remain susceptible to UM. Interestingly, although cancer survivors exhibit a 27% reduced likelihood of childbirth relative to the broader population, those who have battled non-reproductive organ cancers manifest birth rates akin to the general populace [8]. Therefore, it is important to address the reproductive considerations and concerns in this patient demographic.

Existing literature presents a discordant picture regarding the interplay between pregnancy and UM prognosis. Given that pregnancy induces physiological changes in the eye [9], it can be assumed that pregnancy may have an effect on intraocular tumors such as UM. Hartage et al. found that history of pregnancy amplifies the risk of UM [10, 11], suggesting a potential modulation by female hormones on UM's pathophysiology. Other studies have countered this assertion, indicating that prior pregnancies may diminish UM risk [12]. As for prognosis, while some researchers have discerned no notable correlation between pregnancy and UM prognosis [13, 14] Egan et al. contend that a history of childbearing could mitigate the risk of metastatic mortality in intraocular melanoma patients [15]. These various findings make the effect of pregnancy on UM an interesting and worthy aspect of study.

Concerning the ramifications of UM and its therapeutic interventions on pregnancy, the radiative aspect of plaque brachytherapy, albeit localized, has elicited concerns among both patients and clinicians regarding its potential implications for maternal fertility and fetal well-being.

This study aims to evaluate the prognosis of childbearing-aged women who underwent plaque brachytherapy at our institution, and the subsequent pregnancy outcomes.

Materials and methods

Study design and setting

A retrospective, single-center study was conducted using medical records of clinically diagnosed UM patients of childbearing age. These patients were treated with iodine-125 plaque brachytherapy by Dr. Yue-Ming Liu at Beijing Tongren Hospital, Capital Medical University, from January 1, 2012, to December 31, 2019. This study was approved by the ethical committee of Beijing Tongren Hospital of Capital Medical University. Research adhered to the tenets of the Declaration of Helsinki. All adult subjects (Age ≥ 18 years) involved in the clinical data used in this study signed written informed consent. For subjects under 18 years of age, written consent was obtained from their guardians.

Diagnosis and treatment

The patients were diagnosed with UM using ophthalmoscopy, ultrasonography, magnetic resonance imaging (MRI), and fundus fluorescein angiography in accordance with the clinical features of UM. All diagnoses and exclusions of differential diagnoses were made by Prof. Wenbin Wei. Subsequent to diagnosis, all patients immediately underwent plaque brachytherapy (PBT). The size of the plaques was designed by the nuclear medicine department of Beijing Tongren hospital according to the maximum tumor base diameter of each patient. The plaques were surgically placed on the sclera directly over the tumor. A prescription dose of 100 Gy was delivered to the tumor apex. The duration required to achieve this dose was calculated based on the plaque's activity and tumor height. When the prescription dose was reached, the device was removed. Post-PBT, patients were routinely evaluated and were followed until October 2021. Pregnancy outcomes and fetal status were tracked by contacting the attending obstetrician at the hospital of birth.

Data collection

Clinical data of 282 consecutive childbearing-age female patient including age, gender, date of treatment was collected. Tumor diameter was measured by B-scan ultrasonography. For pregnant patients, the date of delivery, pre-pregnancy and post-pregnancy clinical features of the tumor and disease related events, if any, were carefully documented. As all pregnancies culminated in full-term births, conception dates were back-calculated as 40 weeks prior to delivery. In total, 13 women had 13 full-term pregnancies after the diagnosis and PBT treatment of uveal melanoma. Given age's significance as an independent prognostic factor for UM metastases, and a subject pool unsuitable for multivariate analyses, control subjects were chosen based on age similarity, tumor base

Table 1 Basic characteristics of included patients

	Pregnant group (n = 13)	Non-pregnant group (n = 96)	P value
Mean Age	30	31	0.391
Max	38	38	
Min	24	15	
Tumor eye			
Left	4	44	
Right	9	52	
Median BCVA	0.5 (Range 0.03 to 0.7)	0.5 (Range 0.05 to 0.8)	0.495
Tumor Base	12.8±3.5	13.1±2.8	0.176
Tumor Height	7.4±2.9	7.1±2.2	0.416
Median follow-up time for survivors	79 (Range 30 to 114)	61 (Range 21 to 116)	0.094
Metastasis case (%)	2(15.38%)	18(18.75%)	(0.769)
Death case (%)	1(7.69%)	12(12.5%)	(0.616)

diameter, and tumor height, with selections made blind to the patient's survival status.

Statistical analysis

We used a commercially available statistical analysis program (SPSS 25.0; IBM-SPSS Inc., Chicago, IL, USA) for the statistical analysis of the data. Differences between the pregnant and control groups were gauged using independent-sample t-tests and chi-square tests. Analysis-survival comparisons were based on the Kaplan-Meier method, with the endpoint being diagnosis of metastases and death. There were no deaths due to other causes.

Comparability of survival distributions was evaluated using the log-rank test.

Results

Demographics of patients

Ultimately, a total of 109 patients were included in our study with a median age of 32 years (range: 15–38 years), including 13 patients who had pregnancy after PBT treatment for UM (pregnant group, median age 30, range 24–38) and 96 patients were categorized as control subjects (median age 33 range 15–37). Across all participants, tumor heights ranged from 2.4 mm to 14.6 mm (mean: 7.1±2.2) and the mean tumor basal diameter was 13.0±2.9 mm (5.7–20.7 mm) (Table 1). There was no difference in onset age (median, 33 and 30 years old, respectively; $P=0.391$), tumor base diameter (mean, 13.1±2.8 and 12.8±3.5 mm, respectively; $P=0.176$), or tumor height (mean, 7.1±2.2 and 7.4±2.9 mm, respectively; $P=0.416$) between the two groups. All pregnancies were subsequent to the administration of PBT treatment for UM (median 30.3 months, range 4.1–72.3 months). Clinical features and pregnancy outcomes of pregnant patients were shown in Table 2.

Patient outcome

The mean follow-up time was 67.0±27.7 months (median:66.0 months, range:21.0 to 116.0 months). By the study's end, 20 patients had developed metastasis. The location of metastasis was documented for 13 patients: One patient exhibited both bone and liver metastasis,

Table 2 Clinical features and pregnancy outcomes of pregnant patients after PBT (n = 13)

Case No.	Age at Diagnosis	Clinical Features in Primary Survey		Time of PBT (Year/Month/Date)	Pre-Pregnancy Clinical Features		Date of Delivery (Year/Month/Date)	Post-Pregnancy Clinical Features		Outcome
		Base	height		Base	height		Base	height	
1	30	17.0	10.0	2016/10/10	15.8	5.6	2020/2/1	15.8	4.2	Alive
2	28	11.3	5.0	2014/9/22	8.4	3.3	2019/9/1	NA		Alive
3	37	11.3	8.9	2015/4/20	13.0	7.5	2018/3/1	12.9	6.6	Discovered metastasis on 2018/1/1, Dead on 2018/4/1
4	30	19.4	4.4	2019/3/11	12.0	3.6	2020/8/1	NA		Enucleation during pregnancy
5	25	11.4	8.1	2012/3/8	12.0	5.7	2016/12/1	11.6	3.7	Alive
6	31	10.6	6.0	2015/2/2	8.4	6.1	2016/4/1	9.3	3.8	Alive
7	26	11.3	7.7	2013/11/27	NA		2020/9/1	NA		Enucleation in 2014
8	34	17.0	14.6	2016/6/6	NA		2020/5/1	NA		Enucleation in 2017/2
9	32	12.4	6.4	2015/3/23	9.0	3.9	2016/10/1	8.4	3.2	Alive
10	38	7.5	4.9	2015/5/18	7.1	4.2	2018/4/1	7.3	2.8	Alive
11	25	14.7	7.5	2018/3/19	11.9	5.5	2020/10/1	11.2	4.8	Alive
12	26	14.5	8.6	2013/11/6	20.7	8.6	2017/12/1	NA		Enucleation after delivery(tumor body filled vitreous chamber) Discovered metastasis on 2018/1/31
13	24	8.1	4.1	2013/3/1	6.4	0.9	2017/4/1	NA		Alive

while all others developed liver metastasis exclusively. Notably, one was diagnosed liver metastasis two months post-childbirth and 50 months post-treatment; another developed metastasis during pregnancy and passed away one month after delivery. Among the 18 non-pregnant patients, metastasis emerged between 5 months and 6.8 years post-treatment; 12 succumbed to their condition.

In the pregnant group, local recurrence of choroidal melanoma occurred in 2 patients after or during delivery, and 2 other patients developed neovascular glaucoma due to radiation retinopathy. These 4 patients were treated by enucleation immediately or after delivery. Enucleation and pathology examination were conducted at local hospital or our own. All pathology result confirmed the diagnosis of uveal melanoma and revealed necrosis of most of the tumor tissue. None of the other pregnant patients who had examination before and after pregnancy ($n=7$) showed rapid tumor growth or other signs of disease progression. Local tumor recurrence, enucleation and pathological data of the control group were absent from the research.

Surviving women in the study were followed an average of 5.3 years after diagnosis (range, 1.8 years to 9.7 years) and the follow-up pregnancy ranged were from 7 months to 11.1 years (mean, 4.6 years). A total of 13 deaths were documented, and a comprehensive examination of the cases revealed the absence of any other disease that could have potentially contributed to the observed mortality. Furthermore, the presence of metastasis of the tumor was identified in all cases prior to death. Based on these findings, the cause of death in all cases was determined to be related to melanoma metastasis. Metastasis and death rates for the pregnant and non-pregnant groups were 15.38% / 18.75% and 7.69% / 12.5%, respectively. There were no statistically significant differences in metastasis ($P=0.77$) or death rates ($P=0.62$) between the groups.

Considering the limited sample size of pregnant patients, the metastasis-free survival rate and overall survival rate might not be representative for both groups. We compared Metastasis-free survival months and overall survival months of both groups using Kaplan-Meier method (Fig. 1). Pregnant patients and matched control subjects showed no statistical difference in both aspects (hazard ratio (HR): 0.66, 95% confidence interval (CI): 0.15–2.86, $P=0.576$ and HR: 0.48, 95% CI: 0.06–3.66; $P=0.464$). This indicate that patients who chose to become pregnant subsequent to treatment did not exhibit an augmented risk of metastasis or mortality when compared to patients who did not become pregnant.

Pregnancy outcomes

All 13 pregnancies in the study cohort occurred naturally (median: 31 months, range: 3 to 71 months) after PBT

treatment and were full-term, with no reported adverse events. All pregnant patients delivered healthy children and with no report of placenta or infant metastases.

Discussion

This pioneering study centers on the influence of pregnancy on the prognosis of UM patients treated with PBT, specifically within the Chinese population. Furthermore, we are the first to explore whether UM or PBT treatment impacts on the physiological processes of pregnancy and its outcome.

Above all, our results show that the survival and metastasis rates of these patients is similar to that of women of childbearing age who had no pregnancy after treatment. This outcome correlates with previous observations of an absence of estrogen and progestogens [11, 16–18]. Two previous clinical studies [13, 14] also showed no significant adverse effect of post-treatment pregnancy on the prognosis of UM. Similar to our report, both studies treated most of their patients with conservative treatment like proton beam irradiation or brachytherapy. Hence, the parallel prognosis seen in patients, regardless of post-PBT childbirth, could stem from halted mitotic activity and necrosis in a majority of patients post-local radiotherapy [19].

However, isolated cases over the past decades have shown marked UM growth during pregnancy [16, 17]. For instance, Lee et al. documented a pregnant woman whose untreated uveal melanoma enlarged significantly from $8 \times 5 \text{ mm}^2$ to $13 \times 12 \text{ mm}^2$ [17]. Similarly, Yue et al. detailed a Chinese patient with rapid UM expansion during pregnancy, characterized by extensive muscle invasion. Importantly, neither patient received treatment during pregnancy, and in the latter case, only subjective pre-pregnancy patient reports existed. Combining these cases with past studies, we infer that hormonal and ocular changes during pregnancy likely don't affect UM patients that underwent local radiation therapy.

Consistent with previous research, our findings suggest UM does not considerably impact maternal fertility, as seen in numerous reported instances of UM detection during pregnancy or UM patients conceiving [11, 14, 16, 20–22]. Among all these reports, we found only one case report of placental metastasis attributed to choroidal melanoma [23], with no fetal metastases. This suggests an extremely low likelihood of choroidal melanoma metastasizing to the fetus. We also verified that localized radiation via plaque brachytherapy neither significantly affects maternal fertility nor has a teratogenic impact on the fetus. Furthermore, previous studies have demonstrated that the administration of PBT to pregnant women, even without a lead apron protecting the abdomen during treatment, does not result in any significant adverse effects on the fetus [14].

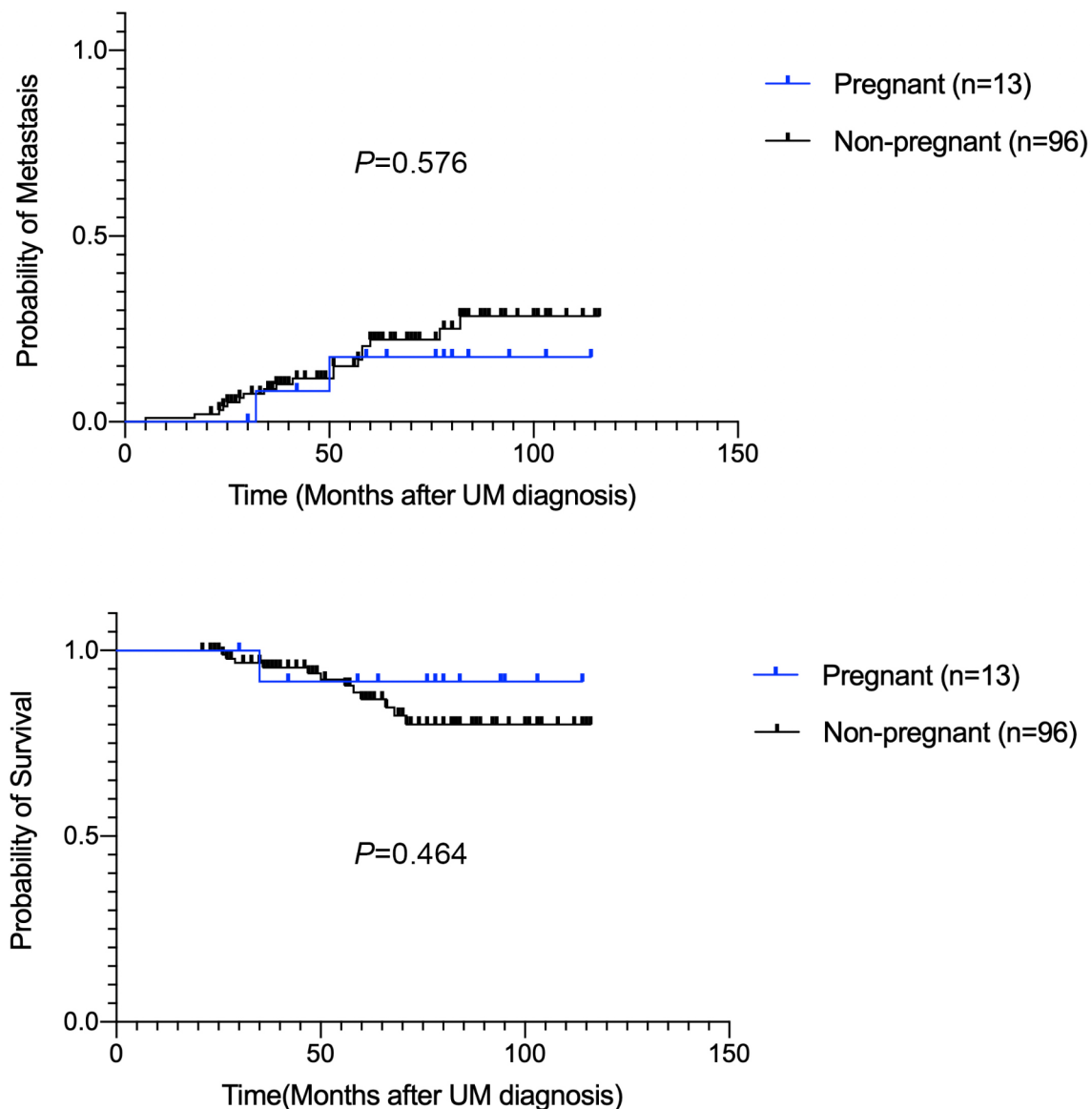


Fig. 1 Kaplan-Meier survival curves for the metastasis (A) and overall survival (B) in the two groups

The following limitations apply to this study: First, as this is a retrospective study, some crucial data could not be accurately answered for statistical analysis, despite our best efforts to collect them. Consequently, details regarding birth weight, route of delivery (vaginal or cesarean) of the pregnancies, and local tumor recurrence, enucleation, and pathological data of the control group were absent from the research. Additionally, as UM primarily affects individuals between the ages of 50 and 70 [1], the number of pregnant cases was limited. It is estimated to

have an incidence rate of four to eight per million, with a reported higher rate among men [2, 24]. Hence, fewer women of childbearing age have a history of uveal melanoma. According to the approximate HR ratio of 0.5 (0.66 and 0.48) in our study, the estimated sample size would be 67 patients who became pregnant and 469 patients in the control group (Log-rank test, Freedman method, $\alpha=0.05$, power=0.8000). Therefore, the current study is inadequately powered. Moreover, although non-reproductive organ cancer survivors exhibit birth rates similar

to the general population [8], many patients abandon childbearing plans upon a malignant tumor diagnosis. These factors have limited the number of cases in studies of pregnancy in UM patients after treatment. Another limitation is the lack of microscopic placenta examinations. We believe that pathological examination of the placenta should be performed as described by previous study [23] to accurately exclude placental metastases. Finally, the study did not include genetic data on the UMs. Therefore, there is no scientific assessment of the malignancy of the tumors.

Over the past two decades, few articles have focused on this issue, and we aspire to pique the interest of researchers in reproductive-age women with malignant tumors. Further clinical and basic studies are needed to draw more definitive conclusions.

Conclusions

While there are ongoing debates and acknowledged limitations surrounding this topic, our study's findings offer encouraging insights for UM patients considering fertility. They also serve as a valuable guide for clinicians when counseling these patients about fertility concerns. We demonstrated that the prognosis of women with uveal melanoma who underwent PBT is not adversely affected by pregnancy. Furthermore, plaque brachytherapy had no significant effect on maternal fertility, nor showed teratogenic effects on the fetus. It is therefore recommended that patients diagnosed with UM who have undergone PBT should not be dissuaded from pursuing pregnancy on the grounds of this medical history. Similarly, if pregnancy occurs, there is no reason to terminate the pregnancy on the grounds of such medical history. However, we strongly recommend consistent ophthalmological monitoring throughout the pregnancy and routine ophthalmic examinations for the child post-birth.

Acknowledgements

Not applicable.

Author contributions

Conception and design of the research: HTW, YML and WBW; Acquisition and interpretation of the data: HTW, YML, WBW and DL, RHZ, WDZ, HYL; Statistical analysis and writing of the manuscript: HTW, YML, DL, RHZ and HYL; Critical revision of the manuscript: HTW and WBW.

Funding

This study was supported by National Natural Science Foundation of China (82220108017, 82141128, 82401283); The Capital Health Research and Development of Special (2020-1-2052); Science & Technology Project of Beijing Municipal Science & Technology Commission (Z201100005520045); Sanming Project of Medicine in Shenzhen (No. SZSM202311018); Scientific Research Common Program of Beijing Municipal Commission of Education (No. KM202410025011); The priming scientific research foundation for the junior researcher in Beijing Tongren Hospital, Capital Medical University (No. 2023-YJJ-ZZL-003).

Data availability

Research data of this study are not publicly available on legal grounds. Data can be made available from the corresponding author on a reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the ethical committee of Beijing Tongren Hospital of Capital Medical University. Research adhered to the tenets of the Declaration of Helsinki. All adult subjects (Age ≥ 18 years old) involved in the clinical data used in this study signed written informed consent. The informed consent of subjects who were less than 18 years old was signed by their guardian.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 22 June 2023 / Accepted: 14 September 2024

Published online: 27 September 2024

References

1. Jager MJ, Shields CL, Cebulla CM, Abdel-Rahman MH, Grossniklaus HE, Stern MH, et al. Uveal melanoma. *Nat Rev Dis Primers*. 2020;6(1):24.
2. Singh AD, Turell ME, Topham AK. Uveal melanoma: trends in incidence, treatment, and survival. *Ophthalmology*. 2011;118(9):1881–5.
3. Shields CL, Shields JA, Cater J, z, KGn, Miyamoto C, Micaily B et al. Plaque Radiotherapy for Uveal Melanoma: Long-term Visual Outcome in 1106 Consecutive Patients. *Arch Ophthalmol*. 2000(118):1219–28.
4. Diener-West M, Earle JD, Fine SL. The COMS Randomized Trial of Iodine 125 Brachytherapy for Choroidal Melanoma, III: initial mortality findings. *AMA Archives Ophthalmol*. 2001(119):969–82.
5. Jampol LM, Moy CS, Murray TG, Reynolds SM, Albert DM, Schachar AP, et al. The COMS Randomized Trial of Iodine 125 Brachytherapy for Choroidal Melanoma: IV. Local Treatment Failure and enucleation in the first 5 years after Brachytherapy. *COMS Rep 19 Ophthalmol*. 2020;127(45):S148–57.
6. Singh N, Bergman L, Seregard S, Singh AD. Uveal melanoma: epidemiologic aspect. Berlin Heidelberg: Springer; 2014.
7. Kaliki S, Shields CL. Uveal melanoma: relatively rare but deadly cancer. *Eye (Lond)*. 2017;31(2):241–57.
8. Hartman M, Liu J, Czene K, Miao H, Chia KS, Salim A, et al. Birth rates among female cancer survivors: a population-based cohort study in Sweden. *Cancer*. 2013;119(10):1892–9.
9. Naderan M. Ocular changes during pregnancy. *J Curr Ophthalmol*. 2018;30(3):202–10.
10. Hartge P, Tucker MA, Shields JA, Augsburger J, Hoover RN, Joseph F, Fraumeni J. Case-Control Study of Female hormones and Eye Melanoma. *cancer Resaearch*. 1989(49):4622–5.
11. Zinkhan M, Stang A, Jockel KH, Marr A, Bornfeld N, Schmidt-Pokrzywniak A. Having children, social characteristics, smoking and the risk of uveal melanoma: a case-control study. *Ophthalmic Epidemiol*. 2013;20(6):360–8.
12. Holly EA, Aston DA, Ahn DK, Kristiansen JJ, Char DH. Uveal Melanoma, Hormonal and Reproductive factors in women. *cancer Resaearch*. 1991(51):1370–2.
13. Egan KM, Walsh SM, Seddon JM, Gragoudas ES. An evaluation of the influence of Reproductive factors on the risk of metastases from Uveal Melanoma. *Ophthalmology*. 1993(100):1160–6.
14. Lemaitre S, Levy-Gabriel C, Desjardins L, Plancher C, Asselain B, Vincent-Salomon A, et al. Choroidal melanoma and pregnancy. *Acta Ophthalmol*. 2016;94(7):e652–60.
15. Egan KM, Quinn JL, Gragoudas ES. Childbearing history Associated with Improved Survival in Choroidal Melanoma. *Arch Ophthalmol*. 1999(117):939–42.
16. Seddon JM, MacLaughlin DT, Albert DM, Gragoudas ES, MF III. Uveal melanoma presenting during pregnancy and the investigation of oestrogen receptors in melanomas. *Br J Ophthalmol*. 1982;66:695–704.

17. Lee CS, Yang WI, Shin KJ, Lee SC. Rapid growth of choroidal melanoma during pregnancy. *Acta Ophthalmol*. 2011;89(3):e290–1.
18. Yue H, Qian J, Ren H, Zhang R, Gan L, Bi YW. Accelerated growth of uveal melanoma related to pregnancy: a case report and literature review. *Int J Ophthalmol*. 2021;14(2):326–9.
19. Shields CL, Shields JA, Karlsson U, Mendum H, Brady LW. Enucleation after Plaque Radiotherapy for Posterior Uveal Melanoma. *Ophthalmology*. 1990;97(12):1665–70.
20. Shields CL, Shields JA Jr., Potter RCE, Mendum PD. Uveal melanoma and pregnancy: a report of 16 cases. *Ophthalmology*. 1991;98(98):1667–73.
21. Miller E, Barnea Y, Gur E, Leshem D, Karin E, Weiss J, et al. Malignant melanoma and pregnancy: second thoughts. *J Plast Reconstr Aesthet Surg*. 2010;63(7):1163–8.
22. Schmidt-Pokrzywniak A, Kalbitz S, Kuss O, Jöckel K-H, Bornfeld N, Stang A. Assessment of the effect of iris colour and having children on 5-year risk of death after diagnosis of uveal melanoma: a follow-up study. *BMC Ophthalmol*. 2014;14:42.
23. Marsh RW, Chu N-M. Placental metastasis from primary ocular melanoma: a case report. *Am J Obstet Gynecol*. 1996(174):1654–5.
24. Park SJ, Oh CM, Kim BW, Woo SJ, Cho H, Park KH. Nationwide Incidence of Ocular Melanoma in South Korea by using the National Cancer Registry Database (1999–2011). *Invest Ophthalmol Vis Sci*. 2015;56(8):4719–24.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.